

Aswin Vattapparambathu Jayaprakash

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An XR enthusiast with hands-on experience in immersive technologies, 3D visualization, and intelligent imaging.

EDUCATION

IMLEX - MSc Engineering, Lighting and XR | MSc in Optics, Image, Computer Vision, Machine Learning and Multimedia | MSc of Computer Science •

University of Eastern Finland | Universite Jean-Monnet | Toyohashi University of Technology

Finland | France | Japan •
09/2024 - 09/2026

- The multidisciplinary IMLEX programme brings together image conversion, lighting and computer science. Students will visit two European countries and Japan during the studies, and graduate from three universities.
- [Upcoming] Master's Thesis – "Human and Societal Augmentation using VR/AR-based Prototyping and AI"
- Host Institution: National Institute of Advanced Industrial Science and Technology (AIST), Japan (Planned, Mar–Aug 2026)
- Focus: Applying XR and AI to enhance human–system interaction and industrial workflows.

BSc (honors) in Physics • Hindu College, University of Delhi

Delhi, India • 11/2020 - 07/2023

WORK EXPERIENCE

XR Research and Development Intern

LIRIS (Laboratoire d'InfoRmatique en Image et Systèmes d'information)

Saint-Étienne, France • 06/2025 - 09/2025

- Enhanced the 6DoF VR Thermal Feedback System (post-IEEE VR 2025) by implementing obstacle-aware thermal rendering in Unity (C#), improving system realism and usability for future XR research.
- Integrated Skinitic Haptic Vest with custom feedback patterns and developed a multimodal XR demo (thermal + haptics+ spatial audio), showcasing how sensory integration can heighten immersion and inspire applications in tele-rehabilitation and human-system interactions.

Student Research Assistant

Hindu College, University of Delhi

Delhi, India • 06/2023 - 01/2024

- Conducted an in-depth review of laser-based headlamp technologies, focusing on advantages such as luminous efficiency, compact design, and adaptive driving beam (ADB) capabilities.

Research Intern

D S Kothari Centre for Research and Innovation in Science Education

Delhi, India • 06/2023 - 07/2023

- Conducted astrophysical analysis of Gaia Early Data Release 3 using Python and SQL to identify extra-tidal star candidates.

Student Research Assistant

Hindu College, University of Delhi

Delhi, India • 06/2022 - 08/2022

- Designed and implemented a low-cost, Arduino-based system using a TCS230 color sensor to measure visible light wavelengths with calibrated precision.

PROJECTS

Face-to-Ball: Deep Learning-based Lighting Transfer (using synthetic data)

03/2025 - 05/2025

Jean Monnet University, France

- Synthesized a dataset of facial images and lighting conditions using 3D modeling tools and Trellis AI.
- Designed and trained a U-Net architecture (with ResNet34 backbone) for relighting tasks based on facial cues.

Real-Time 3D Color Cloud Visualization (WebGL & Three.js)

02/2025 - 02/2025

Jean Monnet University, France

- Developed a real-time interactive color cloud visualization system in Three.js and WebGL
- Designed and implemented GLSL shaders for rendering RGB, CIExY, and CIELAB color spaces
- Integrated VR/MR interactions on Meta Quest, allowing users to manipulate video and color clouds in 3D

Performance and Stress Detection Using Eye Tracking Data

11/2024 - 12/2024

University of Eastern Finland, Finland

- Designed and conducted a study using Tobii Eye Tracker to explore the effects of time pressure on performance and stress.

- Developed gaze analysis workflows using IDT algorithms for saccades and fixation detection.

Path Planning for Robotics

12/2024 - 12/2024

University of Eastern Finland, Finland

- Developed SLAM, autonomous navigation, and odometry in ROS 2 with custom path planning in Gazebo/Rviz.
- Created custom ROS 2 packages and Python nodes for path visualization and navigation control.

Laser Diode Advancements in Automotive Lighting: A Comprehensive Review of Laser-Based Headlamps

06/2023 - 01/2024

Hindu College, University of Delhi, India

- Conducted a detailed review of laser-based headlamps, analyzing advantages like 170 lm/W luminous efficiency, compact design, and adaptive driving beam (ADB) capabilities.
- Identified challenges in heat management, laser safety, and cost reduction to inspire advancements in automotive lighting technologies.

SKILLS

- **Programming languages:** C#, C++, Python
- **Software/Tools:** CUDA, DaVinci Resolve, Latex, OpenMP, PyTorch, ROS2, Scilab (Matlab alternative), Three.js, Unity, WebGL, WebXR
- **Languages:** English (Bilingual), French (Beginner), Hindi (Bilingual), Japanese (Beginner), Malayalam (Native)
- **Soft skills:** Creativity, Critical thinking

CERTIFICATIONS

Python 3.4.3 Training

Spoken Tutorial Project, IIT Bombay

Scilab Training

Spoken Tutorial Project, IIT Bombay

Problem Solving Using Computational Thinking

University of Michigan